AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 28 · STUDENT LESSON

Learning Curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Read the axis. Keep every repeat. Challenge the diagnosis.

Interpret learning-curve patterns as bounded evidence for controlled next experiments.

Keep one habit: Complete size/repeat record → pictured count contrast → three standalone clinic graphs → independent new repeat summaries.

MAIN LESSON · SECTION 1

What does the measured curve suggest,and what remains unknown?

> More training evidence can help without guaranteeing the next improvement.

Learning goals

I can read **training**-size **learning curve**s.

I can describe **variability** across **repeat**ed development runs.

I can avoid unsupported guarantees from a trend.

Recall why validation adds evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Over**fitting**. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.
- What does validation loss add to **training** loss?

End goal

By the end, I can explain **learning curve**s using a traced example and a stated limitation.

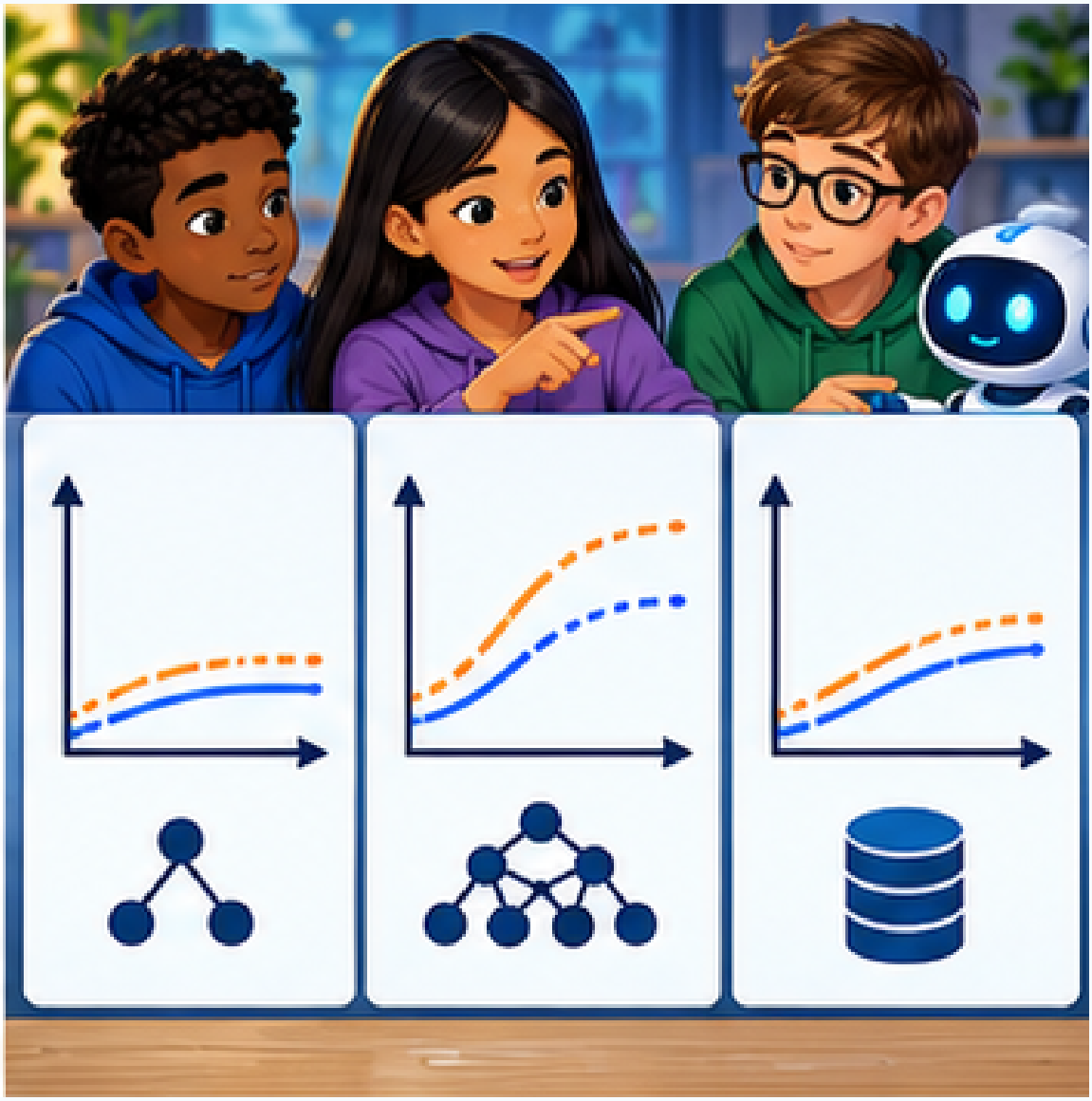
▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Learning curves plot performance against **training** size or effort and help distinguish under**fitting**, over**fitting**, and data limitation.



LEARNING TARGET **Training** sizes 20,40,80 yield validation accuracies 62%,**70%**,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Keep the comparison protocol stable as training size changes

> Training subsets,source groups and all repeated attempts belong in the record.

Learning-curve method and fictional scope

A **learning curve** asks how performance changes as the amount of **training evidence** grows. Keep the validation protocol, model settings and metric comparable.

Use subsets of **training** only; never add validation rows into the **training** subsets. Multiple **repeat**s reveal sensitivity to sample selection or model randomness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

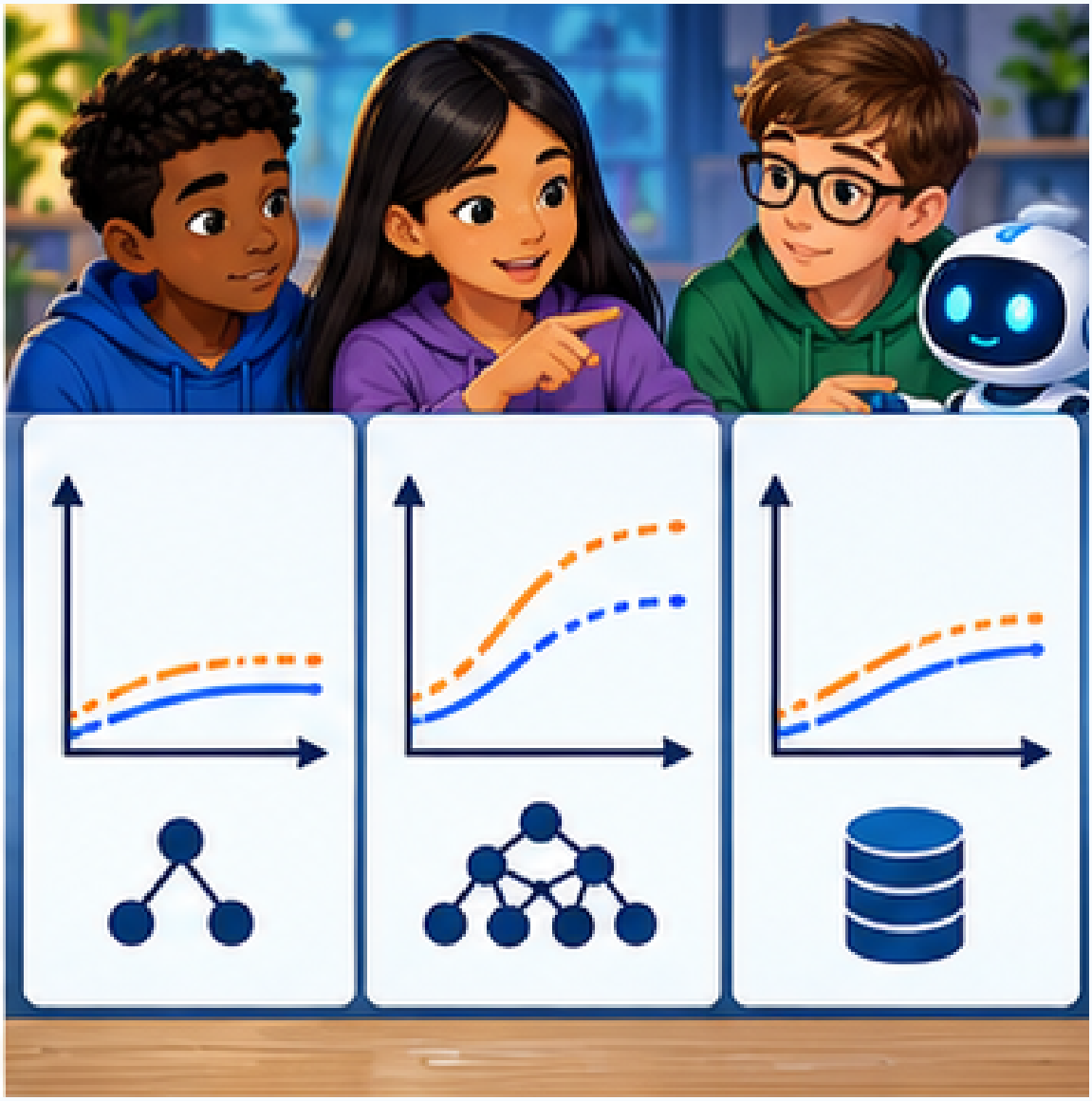
Earlier weeks supplied the model-building **process**; this week examines **learning curve**s and **variability**.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **learning curve**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Trace means and observed ranges before drawing a trend

> A curve summarizes particular runs under a stated protocol.

Complete main size/repeat table

Main supplied training size	Training MAE mean cm	Validation MAEs across 3 repeats cm	Validation mean / min–max / width cm
10	1	7, 9, 8	8 /7–9 /2
20	2	5, 7, 6	6 /5–7 /2
40	3	4, 6, 5	5 /4–6 /2

At **training** sizes 10, 20, 40, validation MAEs across three declared **repeat**s are [7, 9, 8], [5, 7, 6], [4, 6, 5]. Means are 8, 6, 5 and **range**s 2, 2, 2. Mean **training** MAEs are 1, 2, 3.

Validation improves as **training** grows, while **training** error rises because **fitting** more varied examples can be harder.

Explain units,supplied-record status and limits

For the complete curve-clinic figures, we interpret this main numeric-target fixture as height MAE in cm. Training and validation losses are supplied summaries; the model code, full training values and actual source membership are not supplied. Calculating or plotting these summaries is not running model training. The three validation values per size are observed repeats, not a guaranteed future range.

Training loss can rise while validation improves when the larger training set is more varied; this pattern is possible, not a universal rule.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

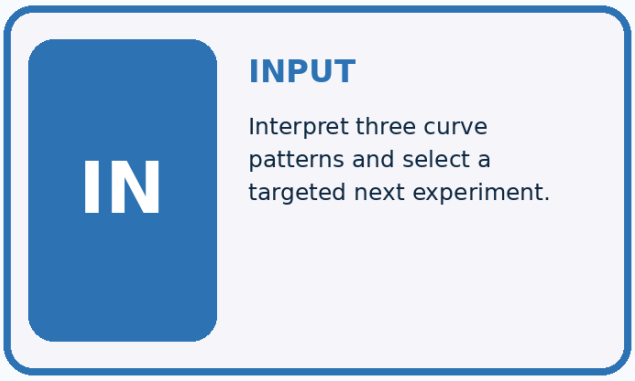
Keep the distinction between an observed result and a proposed next step.

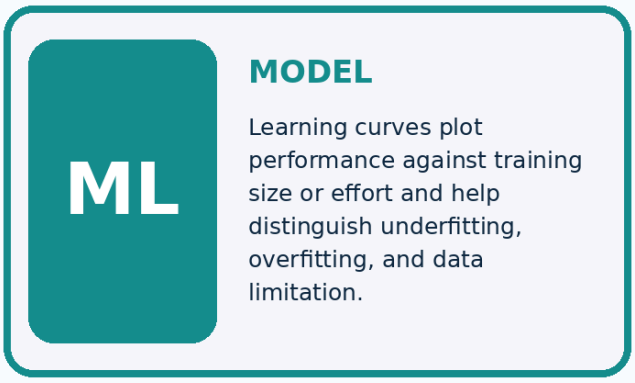
▼ Earlier wording, alternatives and picture transcripts

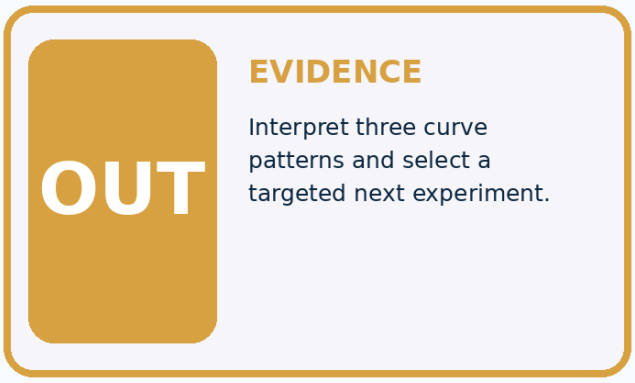
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Interpret three curve patterns and select a targeted next experiment.

MODEL IDEA **Learning curve**s plot performance against **training** size or effort and help distinguish under**fitting**, over**fitting**, and data limitation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Both train and validation scores are low and close: likely under**fitting**. Train high, validation much lower: likely high **variance** or over**fitting**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

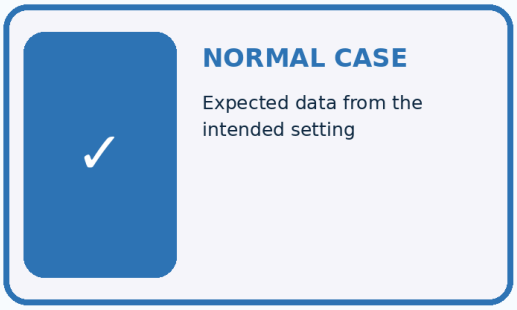
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

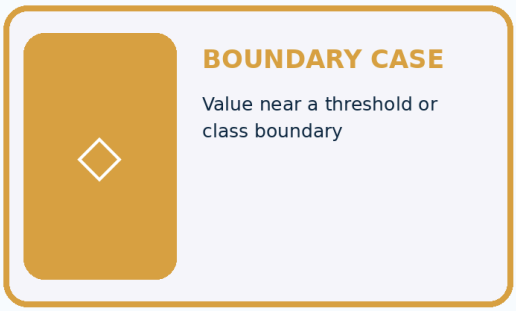
Compare an improving mean with its repeat-to-repeat spread

> A smooth line drawn through a few points is not a promise about the next size.

Four evaluation conditions



NORMAL Expected data and expected behavior.

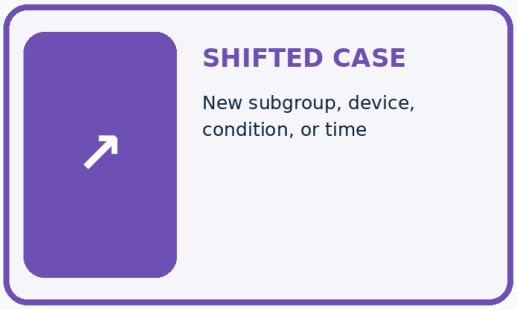


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: A30-plant experiment gives errors [4, 4, 4].

▼ **Check after predicting**

Prediction: Mean remains4; this additional size does not improve the observed mean over20 plants.

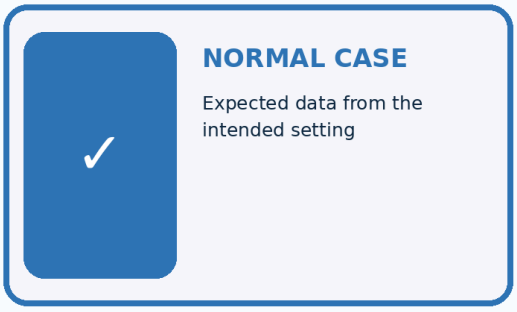
Reason: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

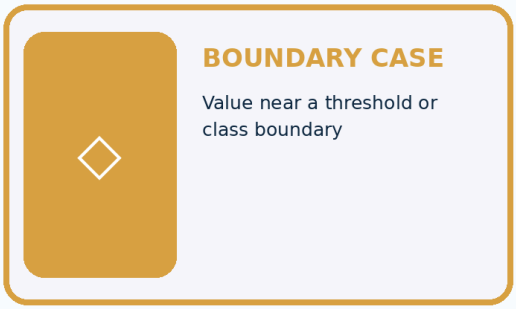
▼ **Earlier wording, alternatives and picture transcripts**

EVALUATION DESIGN

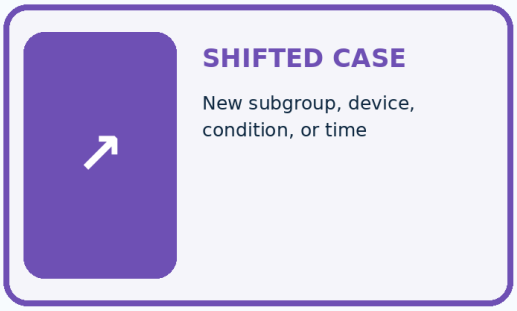
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the quantity and validation denominator beside the picture

> The main loss curve and pictured error-count curve answer different questions.

Complete pictured repeats and separate-fixture scope

Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

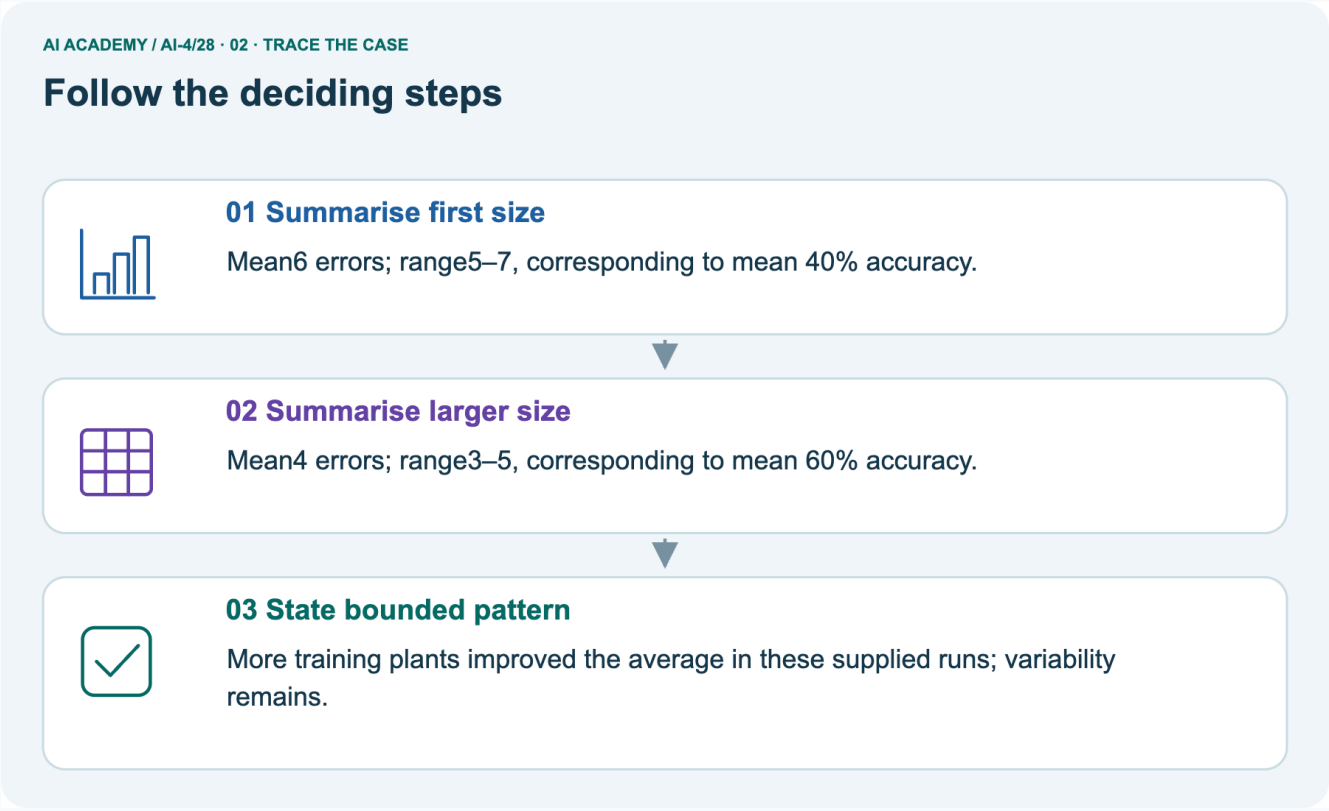
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main fixture is MAE in cm at10/20/40 training rows, with validation means 8/6/5 and widths 2 cm. Picture is classification error counts on one fixed 10-case validation set:at10 training plants errors 6, 5, 7 have mean 6, accuracy 40%; at20 plants4, 5, 3 have mean 4, accuracy

60%. A30-plant set of4, 4, 4 still has mean 4 errors/60%accuracy, but observed width 0 rather than 2 errors.

Do not combine centimetre losses with error counts or treat a zero observed range from three runs as certainty.

Trace means,counts and variability



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Summarise first size	Mean6 errors; range5–7, corresponding to mean 40% accuracy.
2. Summarise larger size	Mean4 errors; range3–5, corresponding to mean 60% accuracy.
3. State bounded pattern	More training plants improved the average in these supplied runs; variability remains.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

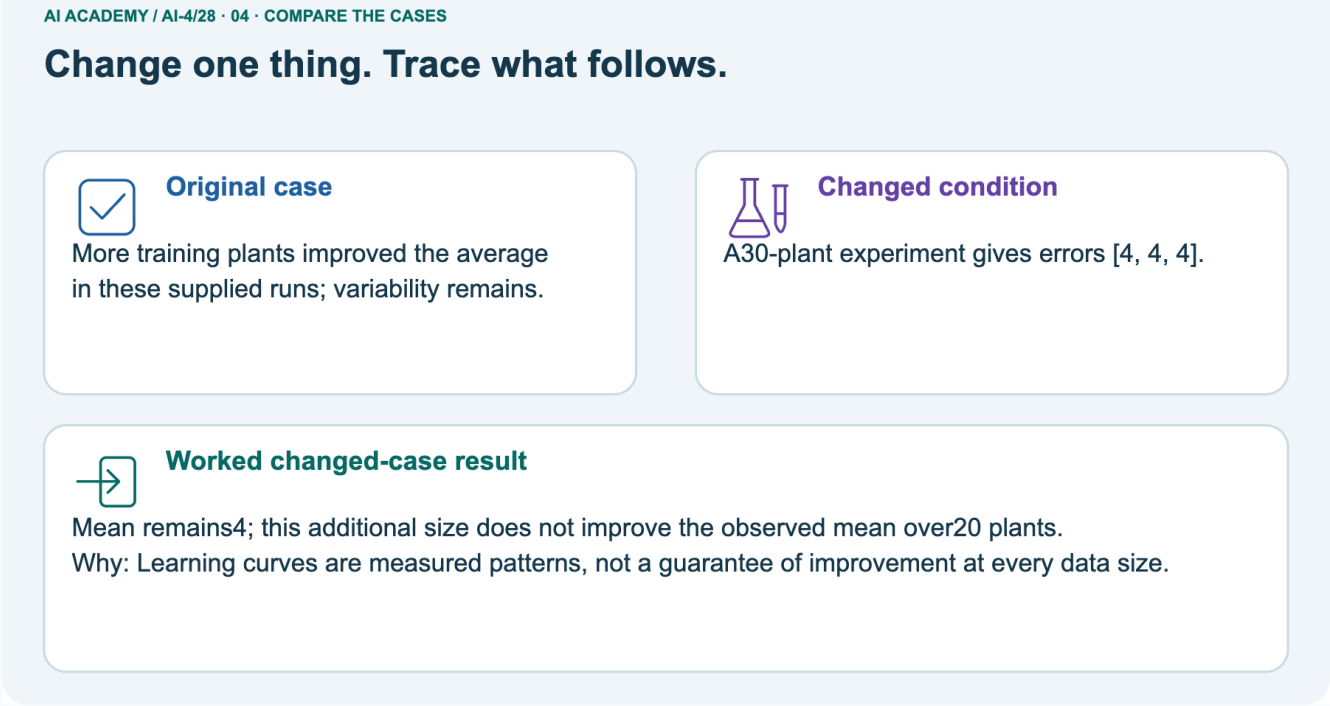
Read the picture in order. More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the30-plant4,4,4repeat set

- Change one thing: A30-plant experiment gives errors [4, 4, 4]. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain cohort and causal limits

- These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

- Read the labelled picture: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to training-rises/validation-falls reasoning

Learning curves and variability at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
10 training rows	Validation 7 9 8	Mean 8 range 2
20 training rows	Validation 5 7 6	Mean 6 range 2
40 training rows	Validation 4 6 5	Mean 5 range 2

The decreasing gap is encouraging in this experiment. It does not establish that doubling again will halve error. A range from three repeats is a descriptive spread, not a confidence interval or a guarantee that future results fall inside it. Repeated runs can share records and therefore are not independent studies.

Why can training error rise while validation error falls? Explain the steps or evidence behind your response.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 28 • WORKED EVIDENCE

1. Both train and validation scores are low and close: likely underfitting.
2. Train high, validation much lower: likely high variance or overfitting.
3. Validation improves as data grows and gap narrows: more representative data may help.
4. Curves suggest hypotheses; confirm with controlled experiments and error slices.

STEP 1 Both train and validation scores are low and close: likely under**fitting**.

STEP 2 Train high, validation much lower: likely high **variance** or over**fitting**.

STEP 3 Validation improves as data grows and gap narrows: more representative data may help.

STEP 4 Curves suggest hypotheses; confirm with controlled experiments and error slices.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

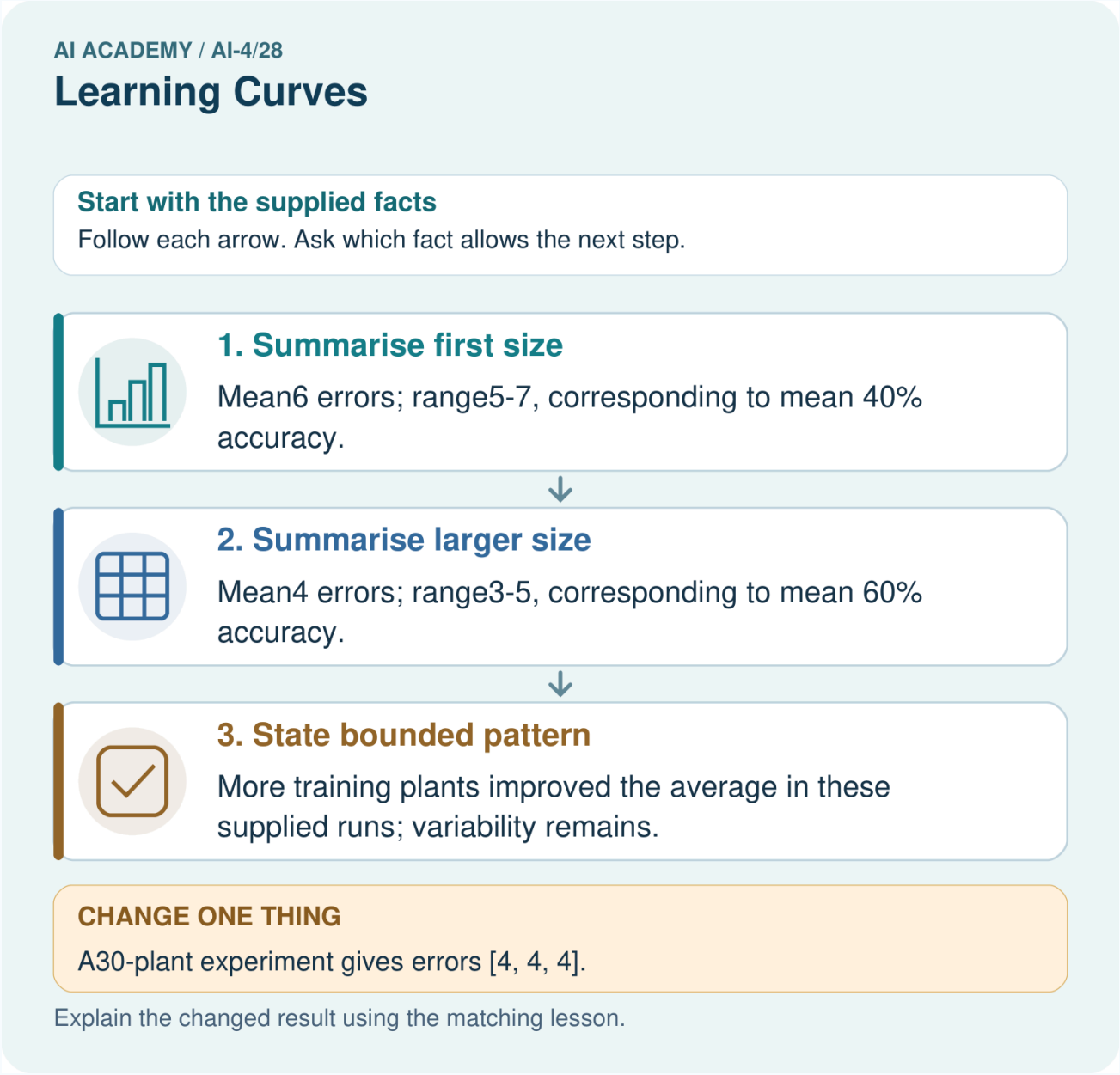
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learning Curves

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a measured trend extended into an unsupported guarantee

> Inspect data quality,source coverage and uncertainty before naming a cause.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Every **learning curve** that rises proves more data will solve the problem.

Explain limits and next-check evidence

- Collect additional data because it addresses a known coverage need, not merely to enlarge a count. When the curve stops improving, inspect labels, features and task suitability.

  Keep the final test reserved while planning these changes.

> **Why is an observed range not a confidence interval?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Every learning curve that rises proves more data will solve the problem. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise a recorded size-and-repeat comparison

> Changed validation cases make a different experiment.

Try supported W2

Use workbook W2 for the connected example "**Learning curve**s and **variability**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Retain subset identities,seeds and evaluation protocol

Record the actual subset identifiers and seeds for every **repeat**.

If the validation population changes at the same time as **training** size, label that as a different experiment rather than attributing the difference only to more **training** data.

Explain the evidence beyond a curve label

- Use the worked case for **Learning curve**s and **variability**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured changed-validation question

- What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent12/24-size summaries

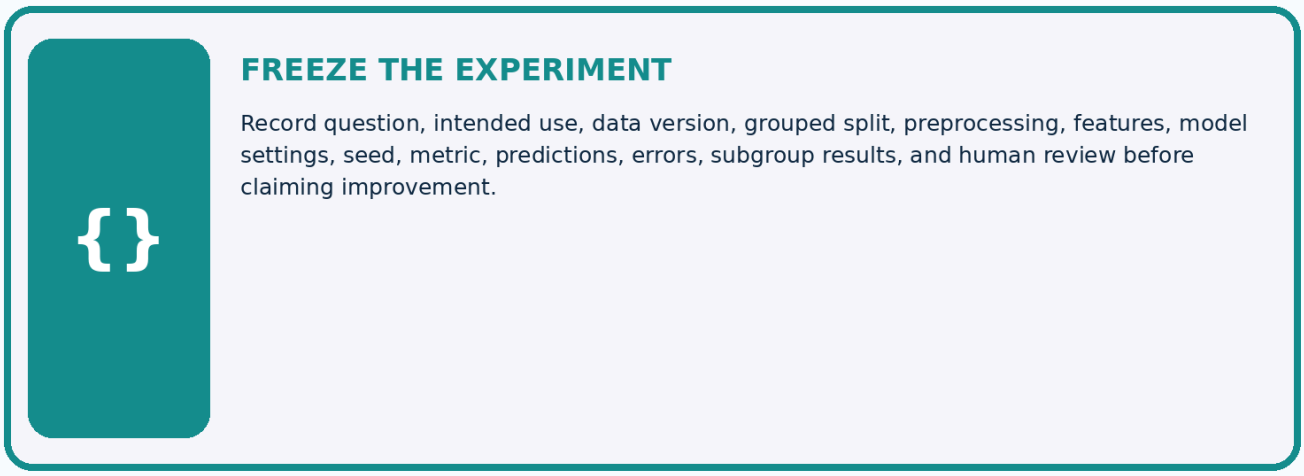
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Read both curves

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run Curve Clinic with three real graphs and all repeat values

> Each pattern supports a hypothesis to investigate,not a definitive diagnosis.

Complete nine-row repeat record

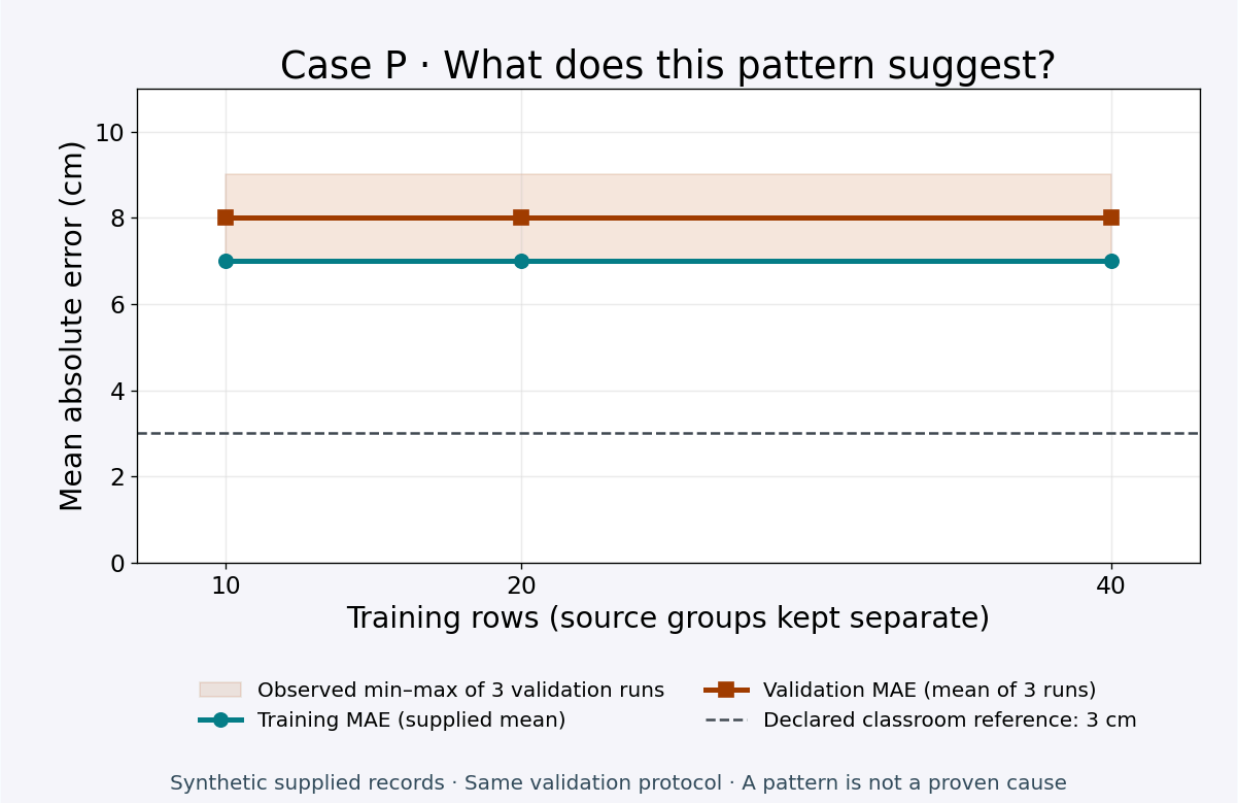
Clinic case	Training size	Supplied training MAE mean cm	All3 validation MAEs cm
P	10	7	7, 9, 8
P	20	7	7, 9, 8
P	40	7	7, 9, 8
Q	10	1	8, 10, 9
Q	20	1	7, 9, 8
Q	40	1	7, 9, 8
R	10	1	7, 9, 8
R	20	2	5, 7, 6
R	40	3	4, 6, 5

Read the axes,reference line and observed band

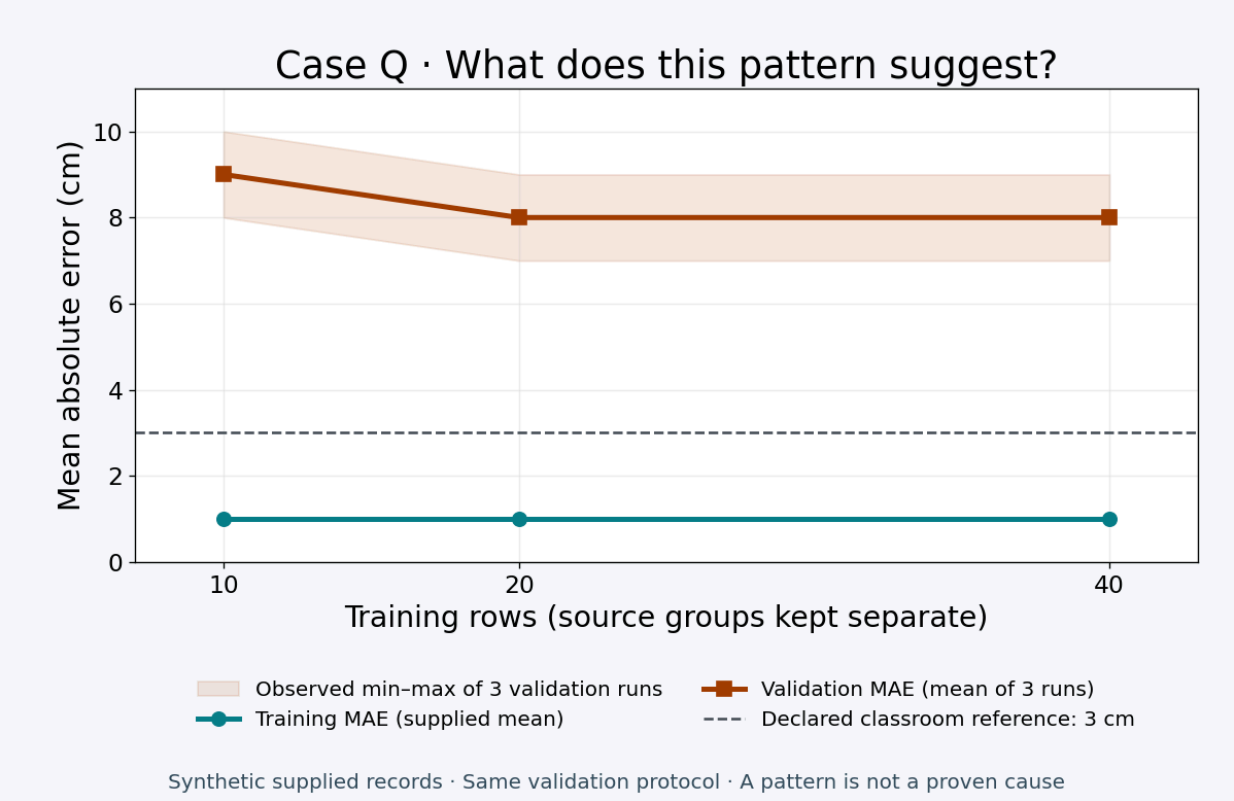
Curve Clinic has three complete synthetic patterns with the same size axis 10/20/40 and height-MAEcm axis. A dashed 3 cmline is a predeclared classroom comparison reference, not a safety or certification threshold. Solid curves show supplied training means and means of the three supplied validation runs. The shaded band is their observed minimum/maximum, not a confidence interval or forecast interval.

Case Rreuses the main summary; P/Qare additional constructed patterns. Model fitting and source memberships are not supplied, so do not claim these graphs establish a causal data-size effect.

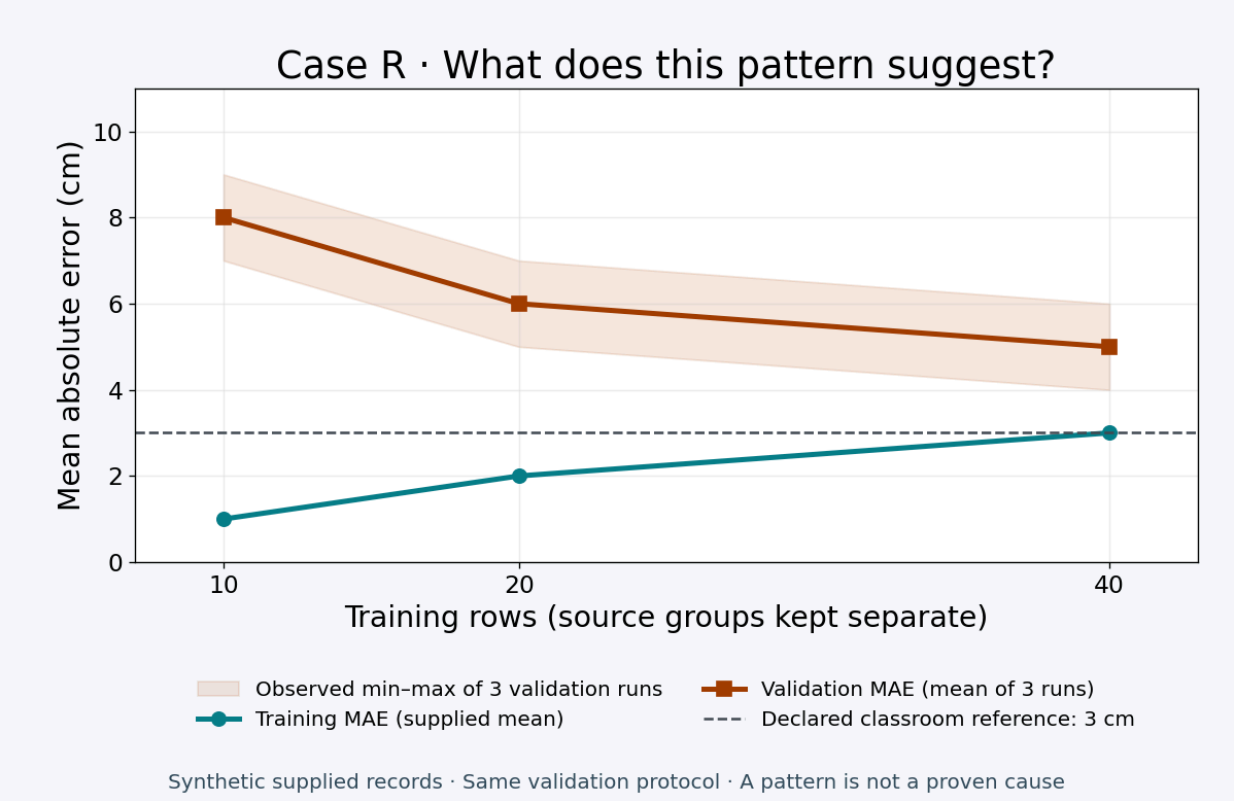
Case P · Read and challenge the pattern



Case Q · Read and challenge the pattern



Case R · Read and challenge the pattern



Match,challenge and plan

1. Match each of the three actual graphs to a possible problem, then challenge your own diagnosis using the supplied table and missing source evidence. Compute means, min/max and widths before reading the reasoning key. State one controlled next experiment for each pattern, including subset/group IDs, seed or no-random-step status, settings, metric and unchanged evaluation context.

   Plotting supplied summaries does not execute a model. Keep the independent 12/24 training-size count sets for an uncoached first attempt.

▼ Check reasoned leads after trying

Case	Possible lead and competing explanation	Small useful next development experiment
P:training 7, validation 8 atall sizes	Both exceed the 3 cmreference and stay close; limited fit is one hypothesis, but labels, features or irreducible variability may also matter.	Check target/label consistency and test one justified feature or candidate change on the same protocol.
Q:training 1, validation 9→8→8	Large gap suggests training-specific fit or a coverage/split problem; size alone has not closed it.	Audit source/label/condition differences, then compare one simpler candidate while keeping data/protocol fixed.
R:training 1→2→3, validation 8→6→5	Gap narrows and observed validation mean improves; further representative data may help, but gains are not guaranteed.	Predeclare an80-row training condition, source-appropriate subsets and three repeats with unchanged validation protocol; record every result, do not predict a guaranteed new mean.

Normal,zero-width,missing-repeat and changed-protocol probes

Complete curve probe	Expected bounded check
Normal R10/20/40 summary	Validation means 8/6/5 cm, widths 2/2/2 cm; training means 1/2/3 cm.
Boundary three repeated losses 4, 4, 4	Observed mean 4, width 0in those runs; not certainty about future runs.
Missing repeat result	Disclose missing/failed run and reason; do not silently replace or selectively omit it.
Changed validation cohort or model settings at the larger size	Not isolated evidence about training quantity; record as a separate changed experiment.

Optional complete supplied-summary program

```
from statistics import mean

cases = {
    'P': ([7, 7, 7], [[7, 9, 8], [7, 9, 8], [7, 9, 8]]),
    'Q': ([1, 1, 1], [[8, 10, 9], [7, 9, 8], [7, 9, 8]]),
    'R': ([1, 2, 3], [[7, 9, 8], [5, 7, 6], [4, 6, 5]])
}
summary = {}
for case, (training_means, validation_repeats) in cases.items():
    summary[case] = []
    for size, training_mae, repeats in zip([10, 20, 40], training_means,
validation_repeats):
        if not repeats:
            raise ValueError('No observed validation mean without repeats')
        result = (size, training_mae, mean(repeats), min(repeats), max(repeats))
        summary[case].append(result)
        print(case, result)
# Supplied synthetic loss records only: no classifier, sampling or fitting is run.
# The observed min/max across three repeats is not a confidence or future interval.
```

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Curve Clinic: match three graphs to diagnoses, challenge each diagnosis, and propose the smallest informative experiment.

1. Write the hypothesis and expected **evidence** before running.

2. Curve Clinic: match three graphs to diagnoses, challenge each diagnosis, and propose the smallest informative experiment.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare new means and repeat ranges

> Keep the training-size axis separate from the validation denominator.

Complete independent repeat sets and scope

At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/**range**s and a careful comparison.

The selected task supplies validation error counts at training sizes 12 and24, with three repeats each. The validation-set size is not supplied, so compare observed count means/ranges without inventing accuracy percentages. A different training size is not the validation denominator.

Describe the observed mean/spread change and the limited three-repeat evidence; do not claim a guaranteed future interval or a causal quantity-only effect.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 28 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

As **training** examples grow from 50 to 100 to 200, validation accuracy is **60%**, 72%, 78%. What does the curve suggest, and what does it not guarantee?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Plot Plant Classifier train/validation performance at several **training** sizes and document the chosen next action.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Training sizes 20,40,80 yield validation accuracies 62%,**70%**,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project curve tied to its experimental protocol

> A next action should name the uncertainty the curve leaves open.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Plot Plant Classifier train/validation performance at several **training** sizes and document the chosen next action.

Save actual training subset/group IDs, size, repeat identity, randomness settings, model/preprocessing version, training/validation metrics and all runs including failures. Keep validation membership and metric comparable; document changes rather than attributing all gains to size. Report observed spread as observed spread and reserve fresh final evaluation for development-informed choices.

A small graph does not guarantee the next data collection will improve performance.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

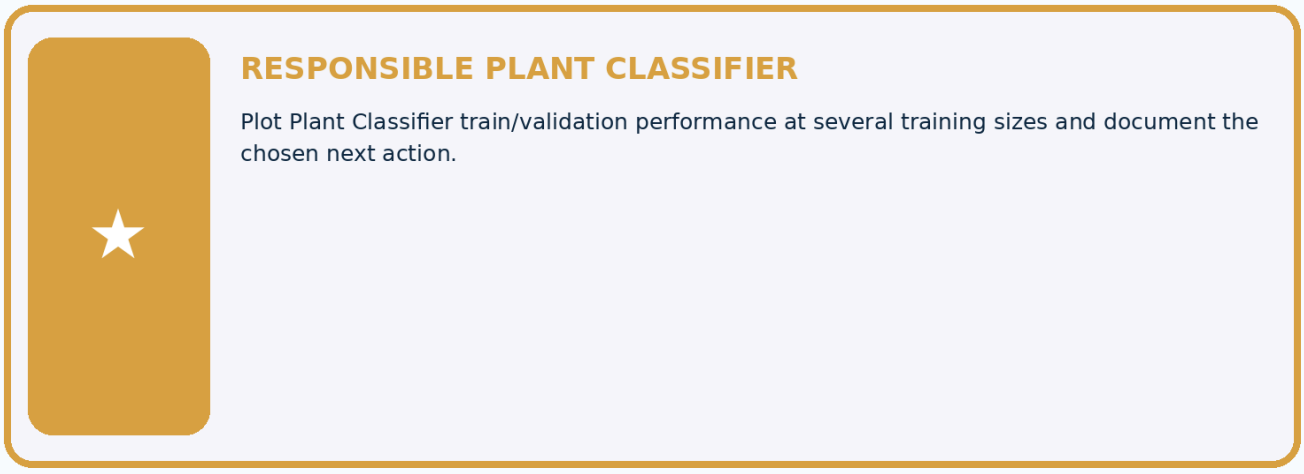
My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with an observed pattern and a testable next step

> Explain which source,protocol or future result the curve does not provide.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Read **training**-size **learning curve**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
learning curve	performance across training sizes	
bias	systematic underfitting	
variance	sensitivity to particular samples	
plateau	little improvement from added data	

Closer-explanation terms

Word	Meaning
Learning curve	Performance plotted against training sample size
Repeat	Another run under a recorded procedure
Variability	How much results differ between runs

Word	Meaning
Range	Maximum minus minimum of observed values

▼ Earlier wording, alternatives and picture transcripts

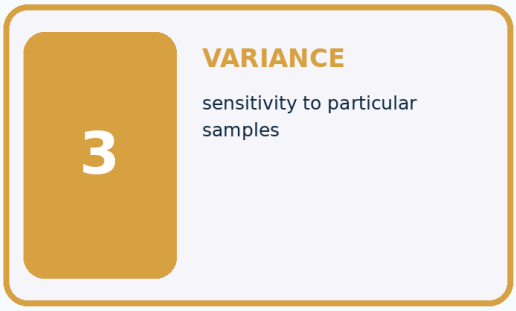
VOCABULARY

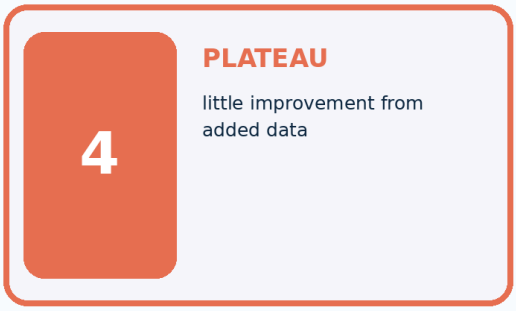
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Every **learning curve** that rises proves more data will solve the problem.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **learning curve**, **bias**, **variance**, **plateau**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Every **learning curve** that rises proves more data will solve the problem.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

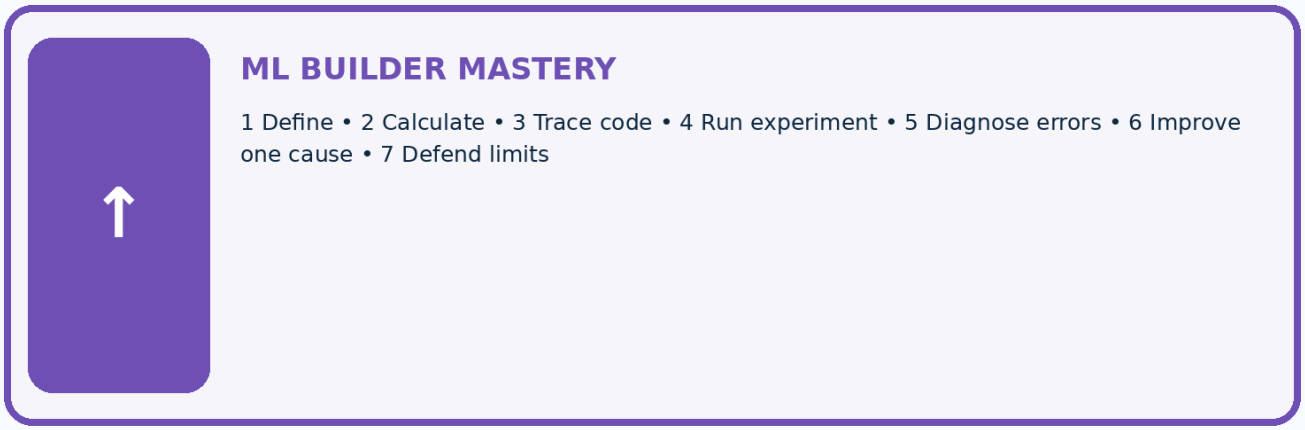
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh12/24-size repeat-count comparison for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Interpret three curve patterns and select a targeted next experiment.

NEXT BRIDGE Carry the experiment **evidence** into: Feature Engineering.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

With20,40,80 **training** leaves, fixed-set validation accuracy is**60%**,**70%**,78%.

Interpret the curve and uncertainty.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

These runs improve with more data. Further representative data may help, but gains are not guaranteed. Controlled **repeat**s reveal **variability**.

Evidence or stage	Value or result
20	60
40	70
80	78

▼ Earlier wording, alternatives and picture transcripts

Week 28 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

More training data on the same validation set

With20,40,80 **training** leaves, fixed-set validation accuracy is**60%**,**70%**,78%.

Interpret the curve and uncertainty.

These runs improve with more data. Further representative data may help, but gains are not guaranteed. Controlled **repeat**s reveal **variability**.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Read training-size learning curves	Explain the deciding step, show a trace or test, and state one limit.
Describe variability across repeated development runs	Explain the deciding step, show a trace or test, and state one limit.
Avoid unsupported guarantees from a trend	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

- Which labelled step matches this detail? More **training** plants improved the average in these supplied runs; **variability** remains.
- 1. State bounded pattern
- 2. Summarise larger size
- 3. Summarise first size
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: State bounded pattern More **training** plants improved the average in these supplied runs; **variability** remains. Read the picture in order.

More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A30-plant experiment gives errors [4, 4, 4]. Which conclusion is supported by the supplied facts?

- 1. Mean remains4; this additional size does not improve the observed mean over20 plants.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Mean remains4; this additional size does not improve the observed mean over20 plants.

Learning curves are measured patterns, not a guarantee of improvement at every data size.

Repair and keep the record

Review Over**fitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3’s evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

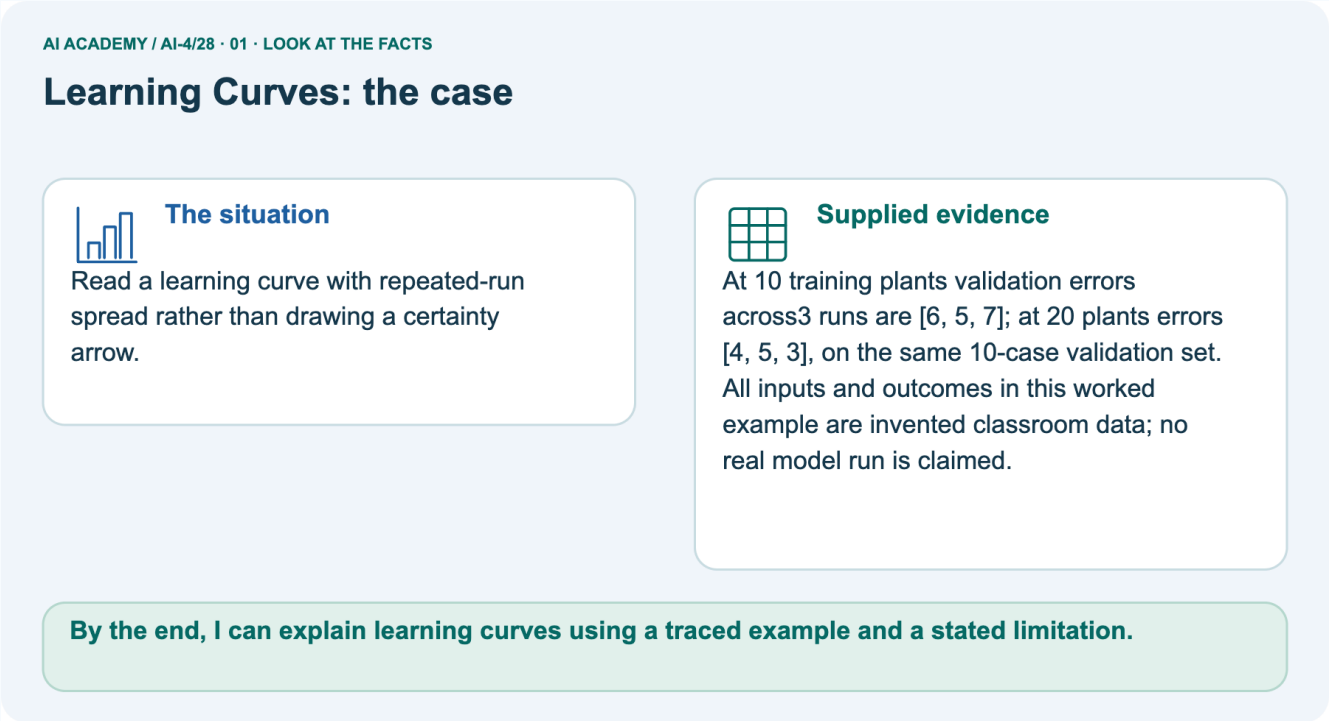
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Learning Curves: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

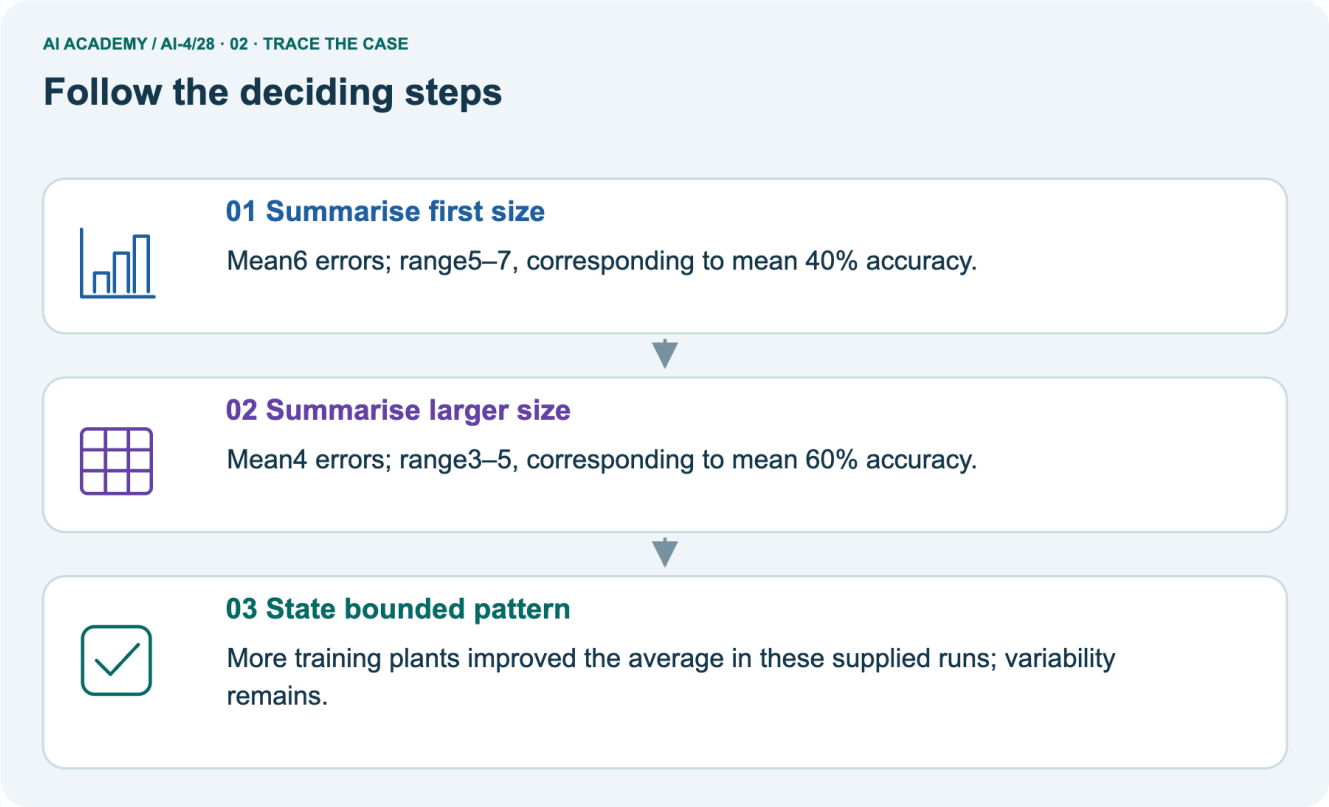
Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

Supplied facts: At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Summarise first size: Mean6 errors; **range**5–7, corresponding to mean 40% accuracy. → Summarise larger size: Mean4 errors; **range**3–5, corresponding to mean **60%** accuracy. → State bounded pattern: More **training** plants improved the average in these supplied runs; **variability** remains.

Point to each numbered stage and explain which recorded **input** it uses.

03. Learning curves and variability at a glance

AI ACADEMY / AI-4/28 · 03 · INSPECT THE LESSON MODEL

Learning curves and variability at a glance

Part	Evidence or action	Interpretation
10 training rows	Validation 7 9 8	Mean 8 range 2
20 training rows	Validation 5 7 6	Mean 6 range 2
40 training rows	Validation 4 6 5	Mean 5 range 2

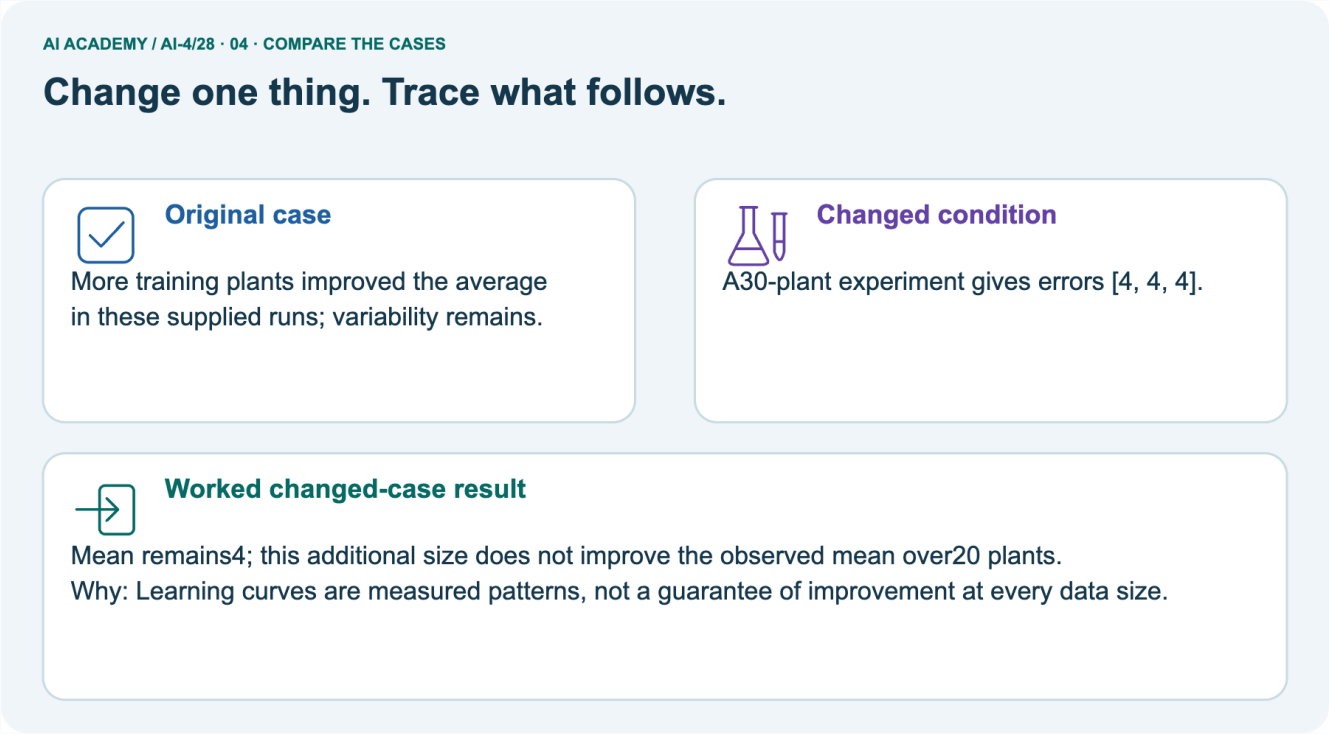
The decreasing gap is encouraging in this experiment. It does not establish that doubling again will halve error. A range from three repeats is a descriptive spread, not a confidence interval or a guarantee that future results fall inside it. Repeated runs can share records and therefore are not independent studies.

Follow the labelled picture with the matching explanation. Open the vector diagram

Learning curves and **variability** at a glance. 10 **training** rows | Validation 7 9 8 | Mean 8 **range** 2. 20 **training** rows | Validation 5 7 6 | Mean 6 **range** 2. 40 **training** rows | Validation 4 6 5 | Mean 5 **range** 2

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



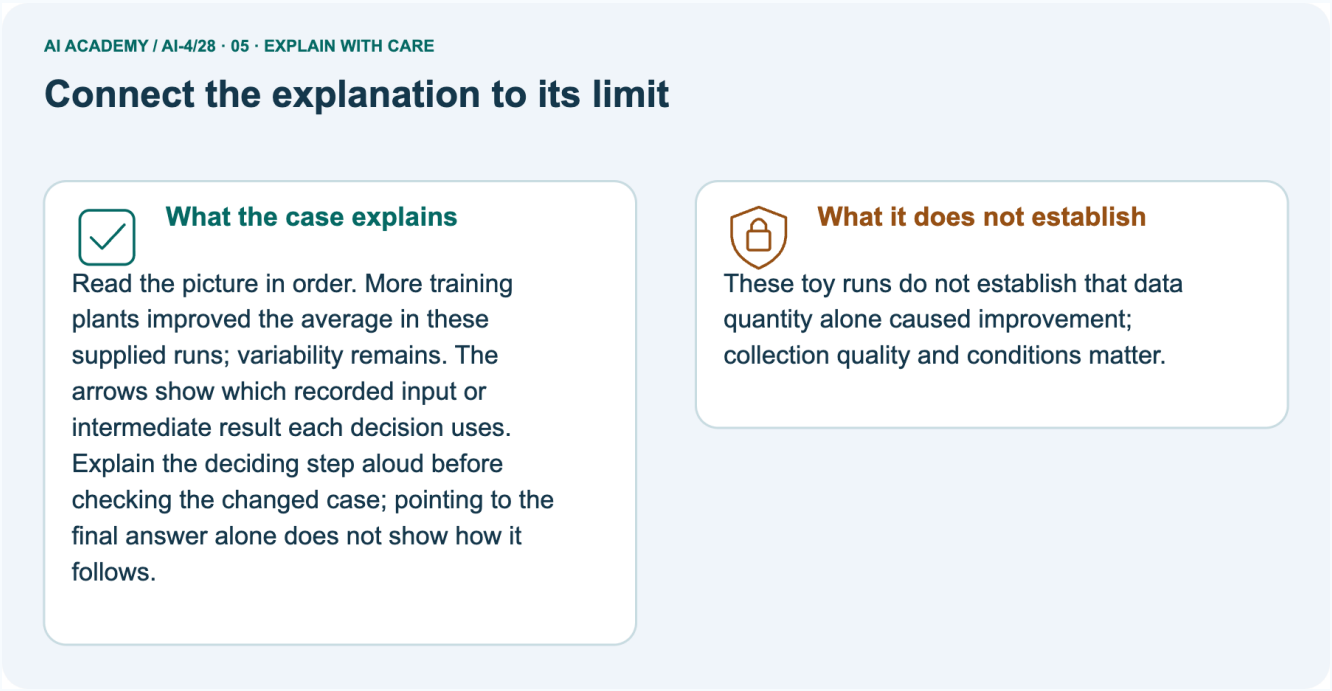
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: More **training** plants improved the average in these supplied runs; **variability** remains. Changed condition: A30-plant experiment gives errors [4, 4, 4].

Expected result: Mean remains4; this additional size does not improve the observed mean over20 plants. Reason: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

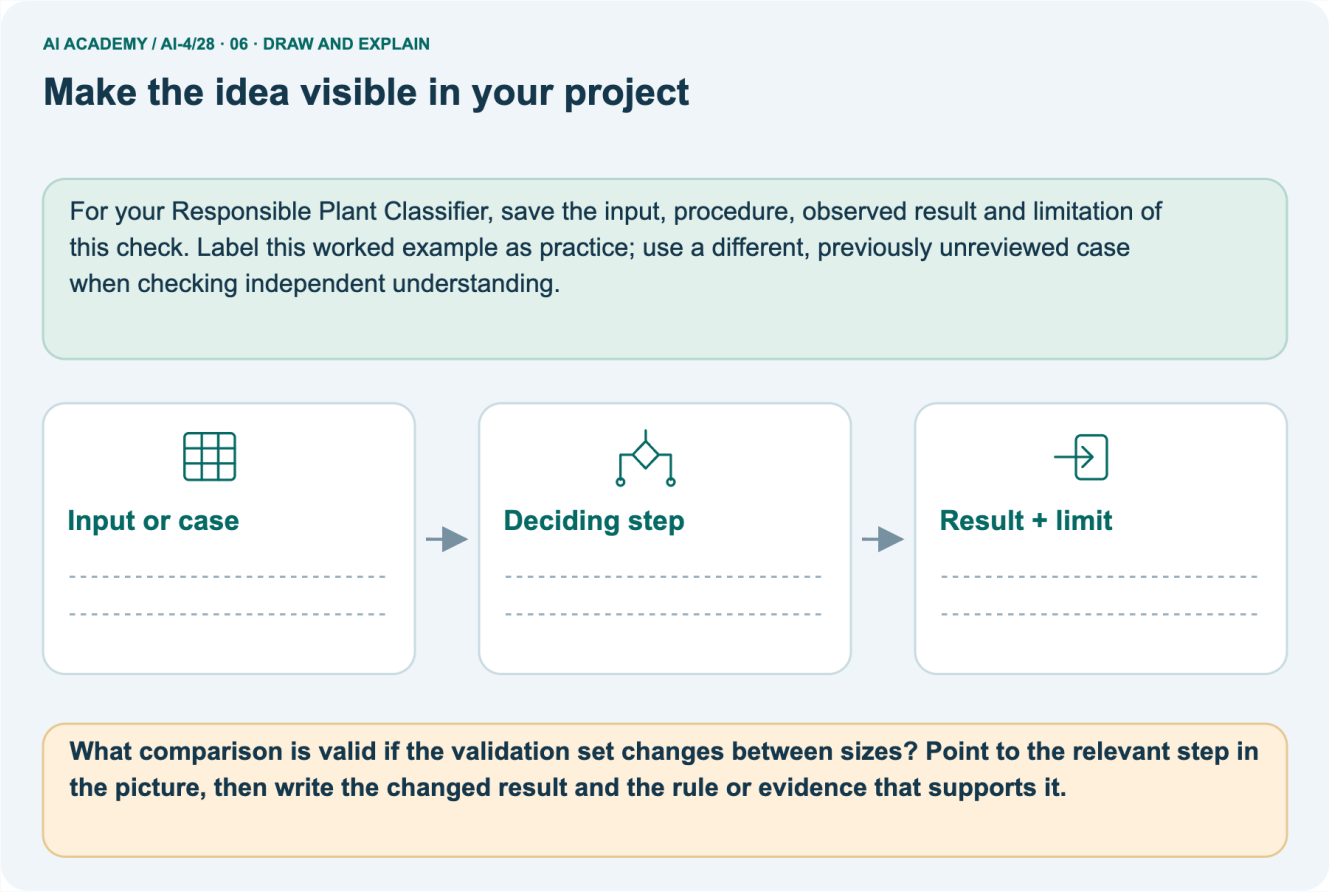
Read the picture in order. More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Learning Curves: the case

02. Follow the deciding steps

03. Learning curves and variability at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.